

Practice Questions for 2024-11-05-NetFlows

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1 Questions

1. In the context of minimum-cost network flow problems, what is the significance of the rank of the incidence matrix A , and how does this relate to the number of independent flow conservation constraints?
 - a. The rank of A is always equal to the number of nodes in the network, implying that all flow conservation constraints are independent.
 - b. The rank of A is always equal to the number of arcs in the network, implying that all flow conservation constraints are independent.
 - c. The rank of A is one less than the number of nodes, indicating that there is one redundant flow conservation constraint.
 - d. The rank of A is one less than the number of arcs, indicating that there is one redundant flow conservation constraint.
 - e. The rank of A is always full rank, regardless of the network structure, implying that all flow conservation constraints are independent.
2. Explain how the complementary slackness condition is applied in the context of minimum-cost network flow problems, and how it relates to the primal and dual solutions.
 - a. Complementary slackness states that for any optimal primal solution, all dual slack variables must be zero.
 - b. Complementary slackness states that for any optimal dual solution, all primal variables must be zero.
 - c. Complementary slackness implies that for each arc, either the primal flow is zero or the dual slack variable is zero (or both).
 - d. Complementary slackness is not applicable to network flow problems because they are always feasible.
 - e. Complementary slackness implies that the sum of primal flows must equal the sum of dual variables.
3. In a minimum-cost network flow problem, how is the initial basic feasible solution typically obtained using a spanning tree, and what is the significance of this approach?
 - a. The initial solution is obtained by setting all arc flows to zero, regardless of the network structure.
 - b. The initial solution is obtained by arbitrarily assigning flows to arcs until flow conservation is satisfied.

- c. The initial solution is obtained by finding a spanning tree and setting flows on arcs in the tree to satisfy flow conservation, while setting flows on arcs outside the tree to zero.
 - d. The initial solution is obtained by solving the dual problem first and then using complementary slackness to find the primal flows.
 - e. The initial solution is obtained by using a random number generator to assign flows to each arc.
4. Suppose you have a minimum-cost network flow problem formulated as a linear program. After solving the dual problem, you find that some dual slack variables are negative. What does this indicate about the current primal solution, and what method could be used to improve it?
- a. The primal solution is optimal because the dual problem is feasible.
 - b. The primal solution is infeasible, and a Phase I method is needed.
 - c. The primal solution is not optimal, and the primal simplex method can be used to improve it.
 - d. The primal solution is unbounded, and the problem has no solution.
 - e. The primal solution is optimal, but the dual solution is not.
5. In a minimum-cost network flow problem, explain the significance of the rank of the incidence matrix A , and how this relates to the number of independent flow conservation constraints.
- a. The rank of A is always equal to the number of nodes, implying all flow conservation constraints are independent.
 - b. The rank of A is always $m - 1$, where m is the number of nodes, indicating one redundant constraint among the flow conservation equations.
 - c. The rank of A is always n , where n is the number of arcs, meaning all flow conservation constraints are independent.
 - d. The rank of A is dependent on the specific network structure and can vary between $m - 1$ and m , where m is the number of nodes.
 - e. The rank of A is irrelevant to the number of independent flow conservation constraints.
6. In a minimum-cost network flow problem, the incidence matrix A has a specific rank. What is the rank of the incidence matrix A , and how does this relate to the number of independent flow conservation constraints?
- a. The rank of A is equal to the number of nodes in the network.
 - b. The rank of A is equal to the number of arcs in the network.
 - c. The rank of A is $m - 1$, where m is the number of nodes, reflecting one redundant constraint due to flow conservation.
 - d. The rank of A is m , where m is the number of nodes, implying all flow conservation constraints are independent.
 - e. The rank of A is $n - 1$, where n is the number of arcs, reflecting one redundant constraint due to flow conservation.
7. In the context of minimum-cost network flow problems, how is the complementary slackness condition applied, and how does it relate to the primal and dual solutions?

- a. Complementary slackness states that for each arc, either the flow is zero or the dual slack is zero, but not both.
 - b. Complementary slackness implies that the sum of primal and dual objective function values is always zero.
 - c. Complementary slackness is only relevant for the dual problem and doesn't directly involve primal variables.
 - d. Complementary slackness requires that all primal and dual variables are strictly positive.
 - e. Complementary slackness is not applicable to network flow problems.
8. In a minimum-cost network flow problem, how is the initial basic feasible solution typically obtained, and what is the significance of this approach?
- a. The initial solution is always the zero flow on all arcs.
 - b. The initial solution is obtained by solving the dual problem first.
 - c. The initial solution is obtained by arbitrarily assigning flows to arcs until flow conservation is satisfied.
 - d. The initial solution is obtained by finding a spanning tree and assigning flows based on supply/demand at leaf nodes.
 - e. An initial basic feasible solution is not necessary for solving minimum-cost network flow problems.
9. Suppose you have solved the dual problem of a minimum-cost network flow problem, and some dual slack variables are negative. What does this indicate about the current primal solution, and what method could be used to improve it?
- a. The primal solution is optimal.
 - b. The primal solution is infeasible.
 - c. The primal solution is feasible but not optimal; the primal simplex method can improve it.
 - d. The primal solution is unbounded.
 - e. The dual problem is infeasible.
10. Consider a minimum-cost network flow problem with m nodes and n arcs. The incidence matrix A is an $m \times n$ matrix. What is the rank of the incidence matrix A , and how does this relate to the number of independent flow conservation constraints?
- a. m
 - b. n
 - c. $m - 1$
 - d. $n - 1$
 - e. $m + n$

2 Answers

1. In the context of minimum-cost network flow problems, what is the significance of the rank of the incidence matrix A , and how does this relate to the number of independent flow conservation constraints?
 - a. **Incorrect.** The sum of rows in A is always zero, meaning there's at least one redundant constraint.
 - b. **Incorrect.** The rank is related to the number of nodes, not arcs.
 - c. The rank of the incidence matrix A is $m - 1$, where m is the number of nodes. This is because the sum of all rows of A is a zero vector, indicating linear dependence among the rows. Therefore, only $m - 1$ of the flow conservation constraints are independent.
 - d. **Incorrect.** The redundancy is related to the number of nodes, not arcs.
 - e. **Incorrect.** The incidence matrix is always singular (non-full rank) for connected networks.
2. Explain how the complementary slackness condition is applied in the context of minimum-cost network flow problems, and how it relates to the primal and dual solutions.
 - a. **Incorrect.** It's not that **all** dual slacks are zero, but rather that for each arc, either the primal flow or the dual slack is zero.
 - b. **Incorrect.** It's not that **all** primal variables are zero, but rather that for each arc, either the primal flow or the dual slack is zero.
 - c. In network flow problems, complementary slackness states that for each arc (i, j) , the product of the primal flow x_{ij} and the dual slack variable z_{ij} is zero: $x_{ij}z_{ij} = 0$. This means that if there is a positive flow on an arc ($x_{ij} > 0$), then the dual slack must be zero ($z_{ij} = 0$), and vice versa. This condition is crucial for verifying the optimality of a solution.
 - d. **Incorrect.** Complementary slackness is a fundamental concept in linear programming and applies to network flow problems as well.
 - e. **Incorrect.** Complementary slackness relates individual primal and dual variables on each arc, not their sums.
3. In a minimum-cost network flow problem, how is the initial basic feasible solution typically obtained using a spanning tree, and what is the significance of this approach?
 - a. **Incorrect.** This is not a feasible solution in general, as it may not satisfy flow conservation.
 - b. **Incorrect.** This approach is not systematic and may not lead to a basic feasible solution.
 - c. A spanning tree is a subgraph that connects all nodes without forming cycles. By selecting a spanning tree, we can set the flows on the arcs in the tree to satisfy the flow conservation constraints at each node. The flows on arcs not in the tree are set to zero. This provides a basic feasible solution, which can then be improved using the simplex method or other optimization algorithms.
 - d. **Incorrect.** While the dual problem can provide information, the initial basic feasible solution is typically found directly in the primal space using a spanning tree.
 - e. **Incorrect.** A random approach is not guaranteed to produce a feasible solution, let alone a basic one.
4. Suppose you have a minimum-cost network flow problem formulated as a linear program. After solving the dual problem, you find that some dual slack variables are negative. What does this indicate about the current primal solution, and what method could be used to improve it?

- a. **Incorrect.** Dual feasibility is a necessary but not sufficient condition for primal optimality. Negative dual slacks indicate that the primal solution can be improved.
 - b. **Incorrect.** Negative dual slacks indicate non-optimality, not infeasibility. A Phase I method is used to find an initial feasible solution, not to improve a non-optimal solution.
 - c. Negative dual slack variables indicate that the current primal solution is not optimal. The primal simplex method can be applied to iteratively improve the solution by moving to adjacent basic feasible solutions until all dual slacks are non-negative (dual feasibility), which guarantees primal optimality.
 - d. **Incorrect.** Negative dual slacks indicate non-optimality, not unboundedness. Unboundedness occurs when the primal objective function can be made arbitrarily small.
 - e. **Incorrect.** If the primal solution is optimal, then the dual solution must also be optimal, and all dual slacks must be non-negative.
5. In a minimum-cost network flow problem, explain the significance of the rank of the incidence matrix A , and how this relates to the number of independent flow conservation constraints.
- a. **Incorrect.** The sum of rows in the incidence matrix is always zero, indicating a linear dependency.
 - b. The rank of the incidence matrix A in a minimum-cost network flow problem is $m - 1$, where m is the number of nodes. This is because the sum of all rows of A is a zero vector, indicating linear dependence among the rows. Therefore, only $m - 1$ of the flow conservation constraints are independent. The remaining constraint is redundant and can be removed without affecting the solution.
 - c. **Incorrect.** The number of arcs (n) is not directly related to the rank of the incidence matrix. The rank is determined by the number of nodes and the network's connectivity.
 - d. **Incorrect.** For a connected network, the rank is always $m - 1$. A disconnected network would have a lower rank.
 - e. **Incorrect.** The rank of A is crucial for determining the number of independent constraints and the structure of the problem.
6. In a minimum-cost network flow problem, the incidence matrix A has a specific rank. What is the rank of the incidence matrix A , and how does this relate to the number of independent flow conservation constraints?
- a. **Incorrect.** While the number of rows in A is equal to the number of nodes, the rows are linearly dependent due to flow conservation, resulting in a lower rank.
 - b. **Incorrect.** The number of columns in A is equal to the number of arcs, but this doesn't determine the rank, which is affected by linear dependence among the rows.
 - c. The rank of the incidence matrix A is $m - 1$, where m is the number of nodes. This is because the sum of all rows of A is a zero vector, indicating linear dependence among the rows. Therefore, only $m - 1$ of the flow conservation constraints are independent.
 - d. **Incorrect.** The flow conservation constraints are not all independent; one is redundant due to the overall balance of supply and demand.
 - e. **Incorrect.** The rank is related to the number of nodes, not the number of arcs. The number of arcs influences the number of columns in A , but not its rank.

7. In the context of minimum-cost network flow problems, how is the complementary slackness condition applied, and how does it relate to the primal and dual solutions?
 - a. Complementary slackness, in the context of minimum-cost network flow problems, states that for each arc (i, j) , either the flow x_{ij} is zero, or the dual slack variable z_{ij} is zero, but not both. This condition is a necessary and sufficient condition for optimality.
 - b. Incorrect. Complementary slackness relates the primal and dual solutions, but it doesn't imply that their objective function values sum to zero. Instead, at optimality, the primal objective function value equals the negative of the dual objective function value.
 - c. Incorrect. Complementary slackness involves both primal (flow) and dual (node potentials and slacks) variables, linking the primal and dual solutions at optimality.
 - d. Incorrect. Complementary slackness allows for primal and dual variables to be zero; it only states that for each arc, the product of the primal flow and the dual slack is zero.
 - e. Incorrect. Complementary slackness is a fundamental concept in linear programming and is directly applicable to network flow problems, providing a necessary and sufficient condition for optimality.
8. In a minimum-cost network flow problem, how is the initial basic feasible solution typically obtained, and what is the significance of this approach?
 - a. Incorrect. The zero flow solution is a feasible solution, but it's not necessarily a basic feasible solution, which is required for efficient solution methods like the simplex method.
 - b. Incorrect. While the dual problem provides valuable information, the initial basic feasible solution is typically found in the primal space using a spanning tree.
 - c. Incorrect. Arbitrarily assigning flows may not lead to a feasible solution, and it certainly won't guarantee a basic feasible solution, which is crucial for efficient optimization.
 - d. The initial basic feasible solution is typically obtained by finding a spanning tree of the network. Starting at the leaf nodes (nodes with only one incident arc), flows are assigned based on the supply/demand at those nodes, and then iteratively working inwards towards the root node. This approach ensures a basic feasible solution (a solution with at most $m - 1$ non-zero flows, where m is the number of nodes) that can be used as a starting point for optimization algorithms.
 - e. Incorrect. An initial basic feasible solution is essential for applying efficient optimization algorithms like the simplex method to solve minimum-cost network flow problems.
9. Suppose you have solved the dual problem of a minimum-cost network flow problem, and some dual slack variables are negative. What does this indicate about the current primal solution, and what method could be used to improve it?
 - a. Incorrect. Negative dual slacks directly contradict the optimality conditions (complementary slackness). A solution is only optimal if all dual slacks are non-negative.
 - b. Incorrect. Negative dual slacks indicate a suboptimal solution, not an infeasible one. The primal solution is assumed to be feasible (satisfying flow conservation and non-negativity constraints).
 - c. Negative dual slack variables indicate that the current primal solution is not optimal. Because negative dual slacks violate dual feasibility, the primal solution, while feasible, can be improved. The primal simplex method can be used to iteratively improve the primal solution by moving to an adjacent basic feasible solution with a lower cost.
 - d. Incorrect. An unbounded primal solution would manifest differently, typically with unbounded dual variables, not just negative slacks.

- e. **Incorrect.** Negative dual slacks indicate a problem with the primal solution, not the dual problem's feasibility. The dual problem is assumed to be feasible.
10. Consider a minimum-cost network flow problem with m nodes and n arcs. The incidence matrix A is an $m \times n$ matrix. What is the rank of the incidence matrix A , and how does this relate to the number of independent flow conservation constraints?
- a. **Incorrect.** The rank is not m because the rows of the incidence matrix are linearly dependent. The sum of all rows is the zero vector.
 - b. **Incorrect.** The rank is not n because the number of arcs (n) is not directly related to the number of independent flow conservation constraints.
 - c. The rank of the incidence matrix A is $m - 1$. This is because the sum of all rows of A is a zero vector, indicating linear dependence among the rows. Therefore, only $m - 1$ rows are linearly independent, representing $m - 1$ independent flow conservation constraints. One constraint is redundant because the total inflow must equal the total outflow in the network.
 - d. **Incorrect.** The rank is not $n - 1$. While $n - 1$ arcs form a spanning tree, the rank is determined by the number of nodes and the linear dependence of the rows.
 - e. **Incorrect.** The rank is not $m + n$. The rank is determined by the linear independence of the rows of the incidence matrix, which is related to the number of nodes, not the sum of nodes and arcs.